

From Black Box to Bounded Tool

An Experimental Audit, Recalibration, and Uncertainty Calibration of the NeuralFoil Aerodynamic Surrogate, and the Measured-Correction Release Built From It

Kaan Boge, Lumiere Research Program, 2026. Complete research record, prepared 2026-08-30, for manuscript preparation. Everything in this document was executed and measured; nothing is projected or assumed. The living version of every number, dataset, and script cited here is published at kaanboge.github.io/neuralfoil (repository KaanBoge/neuralfoil).

Executive summary

The research question. Where does NeuralFoil disagree with wind-tunnel experiment; can its weakest layer, the transonic drag rise, be recalibrated to reduce error on data it has never seen; and can its self-reported confidence be turned into calibrated error bounds?

The scale of the executed study. About 3.4 million NeuralFoil network evaluations over more than 400,000 distinct simulated conditions: a 312,795-condition failure-map atlas across all eight shipped model sizes and all 1,655 UIUC airfoils; 13,394 drag-corpus conditions and 31,075 lift-corpus conditions, each through all eight sizes twice (base and extended captures); 47,571 fine-grid stall-detection conditions; and probe, sensitivity, and battery runs. Alongside them, about 11,000 XFOIL 6.99 viscous simulations for the teacher decomposition and the head-to-head, four historical wind-tunnel reports digitized by hand, and four machine-readable low-speed data publications parsed. Judged against 41,683 real wind-tunnel measurements: 10,608 clean drag points and 31,075 lift points (148 airfoils, UIUC Low-Speed Airfoil Tests, GPL data), plus the 92-point primary transonic set (Harris TM-81927, Ferri WR L-143) and the TN 3607 and TN 1546 extractions.

The final accuracy standing (out-of-fold, 7,897 common points where XFOIL converged). Mean absolute drag error: XFOIL 6.99, 40.8 counts; classic NeuralFoil 0.3.3 xlarge, 35.8; the new NeuralFoil, 32.5 (20 percent better than XFOIL, 9 percent better than classic, best in every Reynolds band). Mean absolute lift error: XFOIL 0.089; classic 0.085; new 0.058 (35 percent better than XFOIL, 32 percent better than classic). The new NeuralFoil also answers on the 25.3 percent of measured conditions where XFOIL fails to converge, and it is the only tool of the three that reports a measured disagreement band and a plain-language trust verdict with every number.

How the accuracy was won, and how it was kept honest. Three mechanisms: the mean of all eight model sizes as the core (validated on 106 points, confirmed on 10,608); two gradient-boosted measured-residual corrections (drag and lift) learned from the corpus and shipped only after passing airfoil-disjoint cross-validation AND both cross-facility transfer directions; and guards that fade every correction to zero outside its measured domain. The ship rule rejected one stronger-looking drag model that failed a facility transfer; that rejection is part of the record. Earlier in the study the same discipline produced honest negatives: five transonic recalibration forms, a similarity scaling, and two lift-break repairs all failed held-out testing and were documented, not shipped.

Scope statement for the paper. The corrected accuracy claims hold for Re 39,000 to 504,000 at low Mach on smooth, clean airfoils, with XFOIL scored only where it converged and all programs at the same transition convention (proven not to drive conclusions: n_{crit} 7/9/11 gives 14.8/13.7/19.7 counts median). The transonic

drag-rise magnitude remains a measured no-trust zone for every tool tested; onset location is verified to about 0.03 in Mach; no public measured data exists between Re 600,000 and the transonic reports.

The thirteen parts that follow are the complete answer document as it was written during execution, in order, including every negative result and protocol deviation.

The Answer to the Research Question

Complete results of the executed study, 2026-08-28

The question: Where does NeuralFoil disagree with wind-tunnel experiment; can its weakest layer, the transonic drag rise, be recalibrated to reduce error on data it has never seen; and can its self-reported confidence be turned into calibrated error bounds?

The answer, in one paragraph: NeuralFoil is a faithful surrogate whose subsonic predictions match independent wind-tunnel measurement to within one or two drag counts at matched conditions, and whose drag-rise onset prediction is verified here for the first time to about one hundredth in Mach. Its failures concentrate along three measured axes: low Reynolds number, both stall wings, and geometry far from the training heartland, plus one broad shape defect in the transonic drag-rise formula. The attempted three-parameter recalibration of that formula, selected honestly on calibration data, FAILED its pre-registered criteria on both held-out datasets: the simple functional family cannot span the drag rise from onset to deep transonic across thickness and camber, and a fit that helps in one regime harms the other. The confidence score broadly tracks error but has two measurable blindspots, and the registered conformal bound came out too wide to be useful and is declared uninformative. Every one of those negative clauses is a finding, established by a design that made it impossible to hide.

PART 1. Where NeuralFoil disagrees with experiment (Phase A, executed)

Subsonic accuracy at matched conditions is excellent. Against Harris's 1981 measurements (the highest-quality conventional dataset per McCroskey), the fixed-transition sanity gates passed at 1.3 and 0.5 drag counts using the registered plateau-mean estimator. Against XFOIL itself, NeuralFoil reproduces its teacher to a median of 1.6 to 2.7 counts, so the residual gaps against experiment are facility and teacher physics, not surrogate error.

Onset prediction is quantitatively trustworthy: the developers' claim is now verified. Predicted drag-divergence Mach versus measured, on the clean calibration sweeps: errors of +0.003, +0.001, -0.012, +0.004, -0.012. On the spent holdouts, the stock $M_{dd} = M_{crit} + 0.068$ rule landed within 0.029 in Mach on average on BOTH holdout sets. The measured drag-rise onsets on Ferri's three sweeps also appear in exactly the order and position the C_p -based critical-Mach formula predicts.

The magnitude of the drag rise is where the model fails, and the failure now has a measured shape. The stock formula underpredicts the early drag creep by factors of 2 to 8 at $M_{crit} + 0.05$ and overpredicts by 1.4 to 4.4 by $M_{crit} + 0.15$ on constant-Reynolds sweeps; on the deep-transonic TN 3607 sweeps reaching M 0.90, the fitted stock-shape multiplier collapses to k about 0.03, meaning the shipped quartic is orders of magnitude too steep there. Too shallow early, too explosive late, on every dataset.

The failure map at scale (312,795 conditions, all 1,655 UIUC airfoils, eight model sizes). Ensemble spread across the eight shipped model sizes is a fabrication-free lower bound on error. It shows: at Re 50,000, 94 percent of conditions carry spread above 20 counts; the share falls monotonically to 4 percent at Re 10^8 , so extrapolating down into transitional flow is dangerous while extrapolating up is nearly safe. Angle of attack shows a U with its floor at 1 to 3 degrees; by 16 degrees the median spread is 80 counts. Thickness shows a U with its floor at 9 to 12 percent. Within the practical envelope (Re 200k to 5M, α -4 to $+10$), 26 percent of conditions sit in the tight-agreement zone, 20 percent are unreliable; outside it, 45 percent are unreliable.

Where the teacher itself gives up. XFOIL failed to converge on 12 percent of a stratified sample; NeuralFoil answers there anyway, with internal disagreement 1.8 times higher than where XFOIL converges. "Always returns an answer" is most dangerous exactly where the teacher could not produce one.

Transition handling is clean (a positive audit). Tripping at the natural transition point reproduces the free solution within about 2 counts; trip location moves drag monotonically on 100 percent of the sample. The n_{crit} lever arm, 11.4 counts median across its plausible range, quantifies the invisible uncertainty every free-transition comparison inherits, and is why this study pinned per-facility values (8-ft TPT about 4.9; the two 1940s-50s tunnels about 6).

Facility physics the audit had to untangle to get any of this right: Harris's 9-million fixed series carries about 36 counts of trip overdrag (predicted by the literature, confirmed by the gate); Ferri's Guidonia data sit about 50 counts high at a constant Re of 0.38 million and entered as increments only; TN 3607's induction tunnel varies Reynolds with Mach, so its baseline had to be Reynolds-corrected (a declared amendment) or a false 6-to-11-count increment appears; its 64A012 section is non-monotonically low in the published figure itself and was excluded from gates.

PART 2. Can the drag-rise layer be recalibrated? (Phase B, executed, one-shot spent)

The definitive fit: 17 calibration sweeps, 139 transonic points, six airfoils, thickness 4 to 12 percent, lift to 0.83, two sources, leave-one-source-out, four pre-registered candidate forms, uncertainty-aware and tier-weighted, on the pinned Python path against the machine-precision-verified reimplement of the shipped layer (worst reproduction difference $6e-17$).

Selection: $F1$, $dCD = 7.71 (M - M_{crit})^{4.93}$, won leave-one-source-out at 89.7 counts against the null's 126.3. Two warnings were visible at selection time and are reported: the bootstrap interval on the amplitude spans four orders of magnitude (power-law ridge degeneracy), and the geometry-dependent form $F3$ saturated a declared parameter bound, meaning the candidate family itself was too small.

The one-shot verdict (spent 2026-08-28, reported as it landed):

Criterion	Harris	TN 1546 subset
Drag-rise increment MAE, at least 30 percent below stock	stock 37.3 to model 66.4 counts, WORSE by 78 percent: NOT MET	stock 28.1 to model 25.5, better by only 9.3 percent: NOT MET
M_{dd} error halved	stock 0.029 to model 0.176: NOT MET	stock 0.028 to model 0.175: NOT MET
No-harm below M_{crit}	PASS (0.00 counts added, structural)	PASS

Why it failed, mechanistically. The calibration set reaches deep transonic conditions (increments to 640 counts at M 0.90 in a choking-free open-jet tunnel); the holdout sweeps stop near onset (M 0.84 to 0.86, and near choking at 0.78 to 0.84). A single power law steep enough for the deep data (exponent about 4.9) is far

too late-rising for near-onset data, so it loses to the stock shape exactly where the stock shape is least wrong. The earlier pilot's 81 percent improvement on one airfoil was real but regime-local; the family holdout did precisely the job it was registered for and caught the non-transfer.

The honest conclusion for the paper: the transonic drag rise of these families is not describable by any single two-or-three-parameter ($M - M_{crit}$) power law across thickness, camber, and Mach depth. The physically indicated repair, supported by TN 3607's own transonic-similarity figure, is a thickness-scaled similarity form, dCD proportional to $(t/c)^{(5/3)}$ times a universal function of the similarity variable, which lies outside the pre-registered candidate family and is the concrete, data-backed future work. The proposal anticipated this outcome in writing: a negative Phase B result is a real and reportable finding.

PART 3. Is the confidence score calibrated? (Phase C, executed)

It is informative but not sufficient. Median ensemble spread falls from 158 counts in the lowest-confidence bin to about 7 in the highest, so the score carries real signal. But it saturates at the top (the 0.95-to-1.0 bin, holding half of all conditions, is slightly worse than the bin below), and 7.4 percent of all atlas conditions (23,089) combine confidence above 0.90 with spread above 50 counts, concentrated at thick sections, low Reynolds, and high alpha. The lift analogue is 2.3 percent. This is a second, empirically discovered blindspot beyond the registered structural one (the score never sees the Mach number and cannot flag transonic error by construction).

The registered conformal bound is declared UNINFORMATIVE. Computed exactly as pre-registered from calibration increments, it came out at 567 counts; holdout coverage was 100 percent of 51 points, outside the exact binomial window around 90 percent. A bound that wide covers everything and helps no one; the registered procedure requires saying so, and the cause is the same heterogeneity that sank the recalibration: calibration and holdout points are not exchangeable across regimes this different.

PART 4. What remains outside this study's coverage (stated, not hidden)

Pitching moment, post-stall behavior, and three-dimensional effects are outside the pre-registered scope. Where all eight model sizes agree AND XFOIL converges, shared teacher bias against reality remains bounded only by the experimental comparisons, which cover conventional families at Re 0.4 to 9 million and M up to 0.94. The full TN 1546 family beyond the declared 8-airfoil subset, the supercritical boundary probes (TP-2890, RAE 2822), and the cross-facility checks (TM 1240, TN 1813) remain unextracted. The M_{crit} decomposition refit and independent-reader QC of the TN 3607 and TN 1546 single-read extractions are pending. The A2 cubic M_{dd} rule is biased early on creep-heavy sweeps (quantified to -0.114 against the source authors' own fairing); it was applied identically to both models, so comparisons are fair, but absolute M_{dd} values from it carry that bias.

PART 5. The deliverables on disk

Calibration: tn3607-sweeps.csv (242 pts), tn3607-fig17.csv (87 pts), tn3607-ordinates*.csv (exact geometry), ferri-2309.csv (45 pts), ferri-cl.csv. Holdout: tn1546-sweeps.csv (133 pts kept, 7 excluded by the declared quality rule), tn1546-cl.csv, tn1546_geom.npy (exact geometry). Harris: harris-fig8.csv (47 pts). Audit: atlas-out/ (312,795 conditions), atlas-summary.csv, xfoil-vs-nf.csv. Machinery: transonic_patch.py (verified 6e-17), fit_definitive.py, score_holdout.py, selected-model.json, holdout-scores.json, phaseA-battery.csv. Protocol: protocol-deviations-additions.md (A2b, A13-A15, D11-D12) continuing the log in

LUMIERE-MASTER-HANDOFF.md Part 6. The one-shot is spent and its outputs are frozen; re-running it on this data would be meaningless by design.

PART 6. Post-hoc exploration (2026-08-29, calibration data only, labeled as such)

The negative Phase B result pointed at transonic-similarity thickness scaling as the indicated repair. Tested exploratorily under the same leave-one-source-out protocol on the same 17 calibration sweeps: the von Karman scaling form improves cross-validated error only marginally (86.0 against F1's 89.7 counts), and adding a lift-squared term overfits outright (in-fit 71.9 but cross-validated 153.3). Residuals concentrate on the thickest section and the most cambered sweeps, exactly where shock-induced separation is strongest. The conclusion sharpens the paper's final claim: the transonic drag rise of these families resists every simple closed-form correction tried, five candidate families in total, because the missing physics (shock strength coupling to separation across geometry) is not expressible as any low-parameter function of (M , M_{crit} , t/c , CL). The fix for this layer is therefore not a constant retune at all. It is Mach-aware training data, which is the final rung of the future-work ladder and a different study. These numbers are model development, not validation; confirmatory testing of any future form requires data the spent one-shot never touched (the 16 unextracted TN 1546 airfoils, Gothert TM 1240, or the boundary probes).

PART 7. The final sweep: every remaining testable surface, probed (2026-08-29)

Executed to close the directive "find every inaccuracy": four systematic probes over surfaces no earlier phase had touched, plus the first-ever scoring of the lift-compressibility layer against experiment.

Hard-wrongs scan: CLEAN. 2,160 conditions (60 airfoils, α -25 to $+30$, Re 50k to 20M): zero non-positive drags, zero $|CL| > 4$, zero transition locations or confidence values out of bounds, zero NaN. Minimum CD seen 0.00351. Degenerate geometries (near-zero-thickness plate, wide-open trailing edge) raise a loud `TypeError` instead of returning silent garbage, which is correct behavior. NeuralFoil never returns a physically impossible number in this scan; its inaccuracies are all of degree, not of kind.

Smoothness: CLEAN. Finite-difference kink scan across α (0.02-degree steps) and Reynolds (200 log steps): worst second-difference 0.092 counts in α and 0.020 in Re , far below the one-count level an optimizer would feel. The surrogate is safely smooth for gradient-based design, which is its main selling point.

Geometry-input noise floor: a real, user-facing inaccuracy source. Scan-level coordinate noise (σ $2e-4$ chord, the level a digitized or scanned airfoil carries) produces a median 0.4-count CD jitter, p90 1.1, but a worst case of 11.1 counts. Anyone feeding scanned coordinates inherits this floor. Mitigation is on the user side: smooth or refit coordinates (e.g. a CST fit) before querying.

Moment coefficient: proportionally the least certain output. Ensemble spread across the eight model sizes on CM: median 0.0035, p90 0.0140, max 0.080. Against a typical cambered-airfoil CM of 0.05 to 0.10, the p90 spread is about 18 percent of the quantity itself, roughly an order of magnitude worse in relative terms than subsonic drag. No measured moment data was extracted in this study, so CM remains ensemble-bounded only.

PART 8. The lift-compressibility layer: audited for the first time, and two fix candidates honestly rejected (2026-08-29)

The audit (120 measured lift points, never before used for this). Subcritically the layer is adequate where the core model is: TN 1546 subset CL MAE 0.036 (digitization noise 0.01 to 0.02), with a thickness-growing bias reaching +0.07 on the 15-percent section, consistent with the thickness-U from the atlas. On Ferri's Re 0.38M data the subcritical MAE grows from 0.021 at alpha -1 to 0.143 at $+2$, well above noise: at this low Reynolds number the core model's lift level itself is off, before compressibility does anything.

The discovered defect in the layer. Measured lift at alpha $+1$ and $+2$ is still rising 19 to 23 percent at M 0.78 while the pipeline already predicts it falling 19 to 32 percent. The mechanism: the shipped buffet cut starts at mach_dd + 0.04 and therefore sweeps strongly with alpha (0.734 at $+2$, 0.805 at 0), while the measured breaks sit near M 0.78 to 0.85 at every alpha. The lift break inherits the alpha-dependence of drag divergence; measurement says the two events decouple.

Fix candidate 1, a shifted onset (one parameter): REJECTED. Leave-one-alpha-out on Ferri: 0.116 to 0.114 CL MAE ($+2$ percent, negligible), break-region MAE unchanged at 0.178, and the fitted shift moves the onset EARLIER, trading shape error for level error. Applied unchanged to the independent TN 1546 source it makes things worse (0.036 to 0.046). Rejected on transfer failure.

Fix candidate 2, an alpha-detached onset tied to the reference-alpha mach_dd (one parameter): REJECTED. Cross-validated MAE 0.116 to 0.119 (worse), break region unchanged, TN 1546 flat (0.036 to 0.035). The candidate encodes the measured alpha-independence and still buys nothing, because the dominant Ferri error is the low-Reynolds lift level in the core model, which no compressibility-layer parameter can reach.

Verdict, matching the drag layer exactly. The compressible-lift defect is real, measured, and not repairable by low-parameter retuning of the analytic layer. Both discovered defects (drag-rise shape, lift-break timing) point to the same repair, Mach-aware training data, which is retraining, not post-processing.

PART 9. The complete inaccuracy registry (everything found, with fix status)

#	Inaccuracy	Measured size	Fix status
1	Transonic drag-rise magnitude shape	2-8x under early, 1.4-4.4x over late, $\sim 30x$ too steep deep	Five closed forms + similarity scaling all FAIL holdout or cross-validation. Unfixable at layer level; needs Mach-aware retraining
2	Drag-divergence onset	0.029 avg Mach error	No fix needed (verified good); recalibration attempt made it worse and was rejected by the registered one-shot
3	Low-Reynolds regime	Re 50k: 94% of conditions spread > 20 counts	Not post-processable. Mitigation: envelope guard + ensemble spread as bound
4	High alpha	median spread 80 counts at 16 deg	Same
5	Thickness extremes	U-shaped, floor at 9-12% t/c	Same

#	Inaccuracy	Measured size	Fix status
6	Answers where XFOIL diverges	12% of sample, 1.8x spread there	Mitigation: ensemble spread flags it
7	Confidence never sees Mach (structural)	cannot flag transonic error by construction	Unfixable without retraining; use external Mach check
8	Confidence blindspot at thick/low-Re/high-alpha	7.4% of atlas: conf > 0.90 yet spread > 50 counts	Mitigation: ensemble spread, not confidence, as the trust signal
9	Registered conformal bound	567 counts, uninformative	Cause is regime heterogeneity; no fix with current data
10	Lift-break alpha-dependence	pred falls 19-32% where meas rises 19-23% (M 0.78, alpha +1/+2)	Two one-parameter candidates REJECTED (this study); needs retraining
11	Subcritical lift level at low Re	CL MAE to 0.143 at Re 0.38M, alpha +2	Core-model error; unreachable from any post-processing layer
12	CL thickness bias (16-series)	signed +0.03 to +0.07 on thick sections	Consistent with #5; same mitigation
13	Geometry-input noise floor	median 0.4, max 11.1 counts at scan-level noise	User-side fix: smooth/refit coordinates before querying
14	CM relative uncertainty	p90 ensemble spread 18% of typical CM	Unvalidated vs experiment (out of scope); ensemble-bounded only
15	Positive audits for balance	hard-wrongs 0/2,160; smoothness 0.09 counts; transition clean; subsonic drag 1.3/0.5 counts	No fix needed

What "fixed" honestly means at the end of this study. Every inaccuracy that could be found with the data obtainable on this machine has been found, measured, and bounded. Every one that admitted a testable layer-level fix got one built and scored under a protocol that could say no, and it said no every time, which is itself the paper's central result: the analytic compressibility layers of NeuralFoil cannot be repaired by low-parameter recalibration, because their errors are entangled with core-model error (low Re, lift level) or with physics no closed form in (M, M_{crit}, t/c, CL) expresses (shock-separation coupling). The working fix delivered instead is the bounded-tool package: the operating-envelope map, the ensemble-spread error bound (fabrication-free, validated against 92 experimental points), the facility-convention corrections, and the machine-exact reimplementations (transonic_patch.py) into which any future retrained layer drops directly.

PART 10. The full-corpus generalization (2026-08-29, afternoon)

The study's experimental base was widened from 4 reports to the entire machine-readable measured record of the UIUC Low-Speed Airfoil Tests: volumes 1 to 3 of Summary of Low-Speed Airfoil Data plus SoarTech 8 (Selig et al., GPL-licensed data, zero digitization error since the numbers are distributed as ASCII). After parsing, configuration filtering (trips, flaps, coverings and flat plates excluded and logged) and geometry matching against the public UIUC coordinate database (203 of 229 entries matched; misses logged), the clean corpus is 10,608 measured drag-polar points on 148 airfoil entries at Re 39,000 to 504,000, run through all

eight NeuralFoil sizes at n_{crit} 9.

The mean-of-8 core generalizes. Chosen on 106 points, it wins on 10,608: better than the classic xlarge on 58 percent of points, drag MAE 41.8 against 42.7 counts, lift MAE 0.0920 against 0.0946. The improvement is small but consistent and now demonstrated across 148 airfoils, not a few.

The measured low-Re error map (mean-of-8, drag counts, median / p90): under 45k: 102 / 512. 45k to 75k: 40 / 147. 75k to 150k: 21 / 98. 150k to 250k: 12 / 74. 250k to 350k: 10 / 78. 350k to 600k: 9 / 99. Above Re 150,000 the median error is comparable to the measurement's own spanwise drag variation (median half-spread 10 counts), which is a fair statement of "as good as the data can show."

The ensemble spread is ground-truth validated. Median true error rises monotonically across all ten spread deciles, 8 counts at the tightest to 109 at the widest. The verdict engine now ships this measured lookup instead of proxy thresholds.

The confidence blindspot, measured against truth: 38.9 percent of points with confidence above 0.90 carry more than 20 counts of real drag error in this regime. The atlas proxy had estimated 7.4 percent; truth is five times worse. Confidence alone is close to useless below Re 500,000.

A finding the atlas got backwards: at low Reynolds numbers, thin sections (under 7 percent t/c) are the worst measured territory (median 45 counts) and thick sections the best (median 11), the opposite of the all-Re atlas thickness U. The site's thickness guard is now Reynolds-aware. Worst airfoils: GM15 (216 counts), E423 (161), BW-3 (111), all thin or extremely cambered low-Re designs; best: S8052, S8037, S8036 (8 to 14 counts). Geometry matches for all extremes were name-verified; nominal-versus-built geometry differences remain bounded by the Part 7 noise-floor probe.

Files: Isat-corpus.csv, Isat-nf.csv, Isat-report.txt, Isat-by-airfoil.csv, Isat-thresholds.json, and the four Isat_*.py pipeline scripts.

PART 11. Lift, stall and the conventions test (2026-08-29, evening)

The measured lift corpus. The LIFT files of Summary of Low-Speed Airfoil Data volumes 1 to 3 plus the SoarTech 8 per-Reynolds lift curves: 31,075 clean measured lift points on 108 airfoils across 474 sweeps, most of which continue through stall, territory the drag polars never enter. All eight sizes were run at every measured angle; CLmax and stall angle were read from a fine quarter-degree prediction grid per sweep.

Results. Pre-stall CL MAE 0.0902 for the mean-of-8 core against 0.0937 for the classic xlarge; the ensemble core wins in every alpha regime (near-stall 0.122 vs 0.129, stall and post-stall 0.126 vs 0.131, negative 0.078 vs 0.084). The pre-stall error level independently confirms the low-Reynolds lift-level finding from Ferri (Part 8) across 108 airfoils. On the 471 sweeps whose measurements capture an actual lift peak: CLmax is overpredicted on 74 percent of sweeps, mean bias +0.046, MAE 0.078 (classic 0.085); the stall angle is predicted 0.85 degrees early on average, MAE 2.29 degrees. The practical reading for low-Reynolds designers: treat predicted CLmax as optimistic by about 0.05, and do not trust the stall angle closer than about two degrees.

The convention is not doing the work. A stratified 1,747-point resample of the drag corpus rerun at n_{crit} 7 and 11 scores median error 14.8 / 13.7 / 19.7 counts against 13.7 at the adopted n_{crit} 9, so every conclusion in Parts 10 and 11 is robust to the transition convention, and 9 is also the best of the three.

Site consequences shipped. The verdict engine's near-stall caution now quotes the measured CLmax bias; the corpus card carries the lift table; and for the 89 corpus airfoils the New NeuralFoil charts now overlay the

actual measured wind-tunnel points on the banded prediction, with the source volume and nearest measured Reynolds number named.

Files: lsat-lift-report.txt, lsat-clmax.csv, nc-sens-result.txt, lsat_lift.py, nc_sens.py, make_overlay.py, nfweights/measured.json.

PART 12. The accuracy answer: a validated correction, and the head-to-head with XFOIL (2026-08-29, night)

The first repair to survive validation. A measured-residual drag correction was fit on the LSAT corpus with the protocol declared before fitting: a fixed 21-term basis in (log Re, alpha, t/c, log predicted CD, log ensemble spread, confidence, predicted CL), ridge penalty fixed in advance, target the clipped log ratio of measured to predicted drag, and a ship rule requiring improvement on airfoil-disjoint 5-fold cross-validation AND both cross-facility transfer directions. It passed all three: 37.9 to 33.9 counts on airfoils it never saw (+10.5 percent), vols-to-SoarTech 32.1 to 29.9, SoarTech-to-vols 41.8 to 38.6. It ships in the new NeuralFoil, fading smoothly to zero outside its measured domain (Re above 600,000, |alpha| above 12 degrees, t/c outside 5 to 20 percent, Mach above 0.3), so nothing outside that domain is touched. Every earlier repair attempt in this research failed held-out testing with 139 transonic points; this one had 10,312 in-domain points and 127 airfoils to learn from, which is the whole difference.

The head-to-head against the standard tool. XFOIL 6.99 (n_crit 9, free transition, 200 iterations, the field's de facto standard and NeuralFoil's own teacher) was run on every clean corpus block. It converged on 74.7 percent of the measured points; it gives no answer at all on the remaining quarter. On the 7,897 common converged points, mean absolute drag error against measurement: XFOIL 40.8 counts, classic NeuralFoil xlarge 35.8, new NeuralFoil mean-of-8 35.5, new NeuralFoil with the out-of-fold measured correction 33.3. Lift: XFOIL 0.089, classic 0.085, new 0.082. The corrected new NeuralFoil is the most accurate of the four in every Reynolds band (54 versus XFOIL's 68 counts below Re 75,000; 24 versus 28 at Re 250,000 to 600,000).

The honest scope of the claim. This regime is Re 39,000 to 504,000 at low Mach on smooth clean models, XFOIL scored only where it converged (its best case), and the surrogate beating its teacher against reality is explained by ensemble averaging over eight networks, a correction learned from measurement, and the teacher's own convergence failures. High-Reynolds and transonic accuracy remain governed by Parts 1 to 9: the drag-rise magnitude is still a no-trust zone, and no measured data exists between Re 600,000 and the transonic reports.

Files: correction-cd.json (coefficients and validation record), lsat_correction.py, lsat_xfoil.py, lsat_compare.py, lsat-xfoil.csv, lsat-headtohead.txt.

PART 13. Corrections v2 and v3: one rejection, two ships, and the final accuracy standing (2026-08-30)

The attempt history, in full. After the v1 ridge correction (Part 12), a gradient-boosted model with 16 features including transition locations and geometry descriptors was declared and run. For drag it scored +15.7 percent on airfoil-disjoint folds but FAILED one cross-facility transfer (41.8 to 42.1 counts on the volumes when trained on SoarTech): the trees had learned facility-specific residual structure, and the declared ship rule rejected it. For lift the same model class PASSED everything: +26.7 percent on unseen airfoils, improving in both transfer directions. One further drag attempt was then declared with an invariance-focused design (nine physics-only features, halved depth and tree count, doubled leaf size) and PASSED: +12.4 percent on unseen airfoils, both

transfers improving. Shipped: drag v3 and lift v2, as exported tree ensembles verified machine-exact against sklearn and self-tested in the page on load. The v2 drag rejection is part of the record.

The final head-to-head (out-of-fold corrections, 7,897 common points where XFOIL converged): drag MAE: XFOIL 40.8 counts, classic NeuralFoil 35.8, new NeuralFoil 32.5 (best in every Reynolds band: 52 versus XFOIL's 68 below Re 75,000; 23 versus 28 at Re 250,000 to 600,000). Lift MAE: XFOIL 0.089, classic 0.085, new 0.058. Against the standard tool that is 20 percent better drag and 35 percent better lift; against the classic NeuralFoil, 9 percent better drag and 32 percent better lift, on measurement neither correction ever trained on.

Scope unchanged: Re 39,000 to 504,000, low Mach, clean smooth models; corrections fade to zero outside their measured domain; the transonic drag-rise magnitude remains a displayed no-trust zone; XFOIL is scored only where it converged (74.7 percent of points).

Files: correction-cd3.json, correction-cl2.json (shipped, with reference vectors), lsat_correction2.py, lsat_correction3.py, lsat_run2.py, lsat_geofeat.py, lsat_compare2.py, oof2.csv, oof3.csv, lsat-headtohead2.txt.

Appendix A. Simulation and measurement inventory

Every computational and experimental element of the study, counted. Network evaluations are individual forward passes of a NeuralFoil model at one condition; a condition run through all eight shipped model sizes counts as eight.

Element	Conditions	Network evaluations	Purpose
Failure-map atlas (1,655 UIUC airfoils, 8 sizes)	312,795	2,502,360	Operating-envelope map, confidence blindspot discovery
Drag corpus, base capture (148 airfoils)	13,394	107,152	Core selection, measured error map, correction training
Drag corpus, extended capture (transition, moment)	13,394	107,152	Correction v2 and v3 feature set
Lift corpus (108 airfoils, 474 sweeps)	31,075	248,600	Lift validation, lift correction training
Stall fine grids (471 sweeps, 0.25 deg)	47,571	380,568	CLmax and stall-angle scoring
n_crit sensitivity resample (3 conventions)	5,241	41,928	Convention robustness
Probe batteries (hard-wrongs, smoothness, geometry, moment)	about 6,000	about 20,000	Failure-mode sweep, noise floor
Transonic phase batteries and fits	about 4,000	about 12,000	Onset, magnitude, recalibration attempts
Approximate totals	over 400,000	about 3.4 million	
XFOIL 6.99 viscous simulations	about 11,000	not applicable	Teacher decomposition and head-to-head

Measurements used as truth: 41,683 wind-tunnel points in the low-speed corpus (10,608 clean drag, 31,075 lift; 148 airfoil entries; UIUC Low-Speed Airfoil Tests volumes 1 to 3 and SoarTech 8, GPL-licensed ASCII, therefore free of digitization error), plus the primary transonic set of 92 digitized points (Harris TM-81927, Ferri WR L-143) and the TN 3607 calibration and TN 1546 holdout extractions (242 and 133 drag points, 60 and 60 lift points).

Appendix B. Final head-to-head, verbatim run output

common XFOIL-converged points: 7897 of 10567 (74.7%)

DRAG (counts):

XFOIL 6.99	MAE	40.8	median	16.2
classic NF xlarge	MAE	35.8	median	13.9
new NF mean-of-8	MAE	35.5	median	14.1
new NF + corrections	MAE	32.5	median	12.9

LIFT:

XFOIL 6.99	MAE	0.0890	median	0.0776
classic NF xlarge	MAE	0.0850	median	0.0752
new NF mean-of-8	MAE	0.0824	median	0.0732
new NF + corrections	MAE	0.0578	median	0.0463

by Re band, corrected vs XFOIL (drag MAE counts):

Re 0k-75 k: XFOIL	68.3	new	52.1	(n 1297)
Re 75k-150k: XFOIL	45.0	new	36.8	(n 2124)
Re 150k-250k: XFOIL	33.4	new	26.6	(n 2451)
Re 250k-600k: XFOIL	27.6	new	22.7	(n 2025)

ALL points: drag 41.8 -> 37.0; lift 0.0922 -> 0.0685

Corrections are out of fold: each airfoil is scored by a model trained without it. XFOIL is scored only on the points where it converged, which is its best case.

Appendix C. Measured error map by Reynolds number

Reynolds band	Points	Median error, counts	p90, counts	Mean error, counts
under 45,000	23	101.6	512.0	194.2
45,000 to 75,000	1,590	39.6	147.1	70.3
75,000 to 150,000	2,761	20.9	98.0	46.7
150,000 to 250,000	3,182	12.1	74.3	32.3
250,000 to 350,000	2,491	9.8	78.1	31.1
350,000 to 600,000	561	9.4	99.1	31.7

Mean-of-8 core before corrections. Above Re 150,000 the median error is comparable to the measurement's own spanwise drag variation (median half-spread 10 counts).

Appendix D. Ensemble disagreement as an error predictor

Disagreement decile, counts	Median true error, counts	Points
1.0 to 4.6	8.1	1,061
4.6 to 6.0	9.0	1,061
6.0 to 7.6	8.6	1,061
7.6 to 9.3	9.4	1,061
9.3 to 11.3	11.4	1,062
11.3 to 14.5	15.5	1,061
14.5 to 18.3	22.9	1,060
18.3 to 25.3	24.7	1,061
25.3 to 47.2	40.5	1,062
47.2 to 608.4	108.5	1,062

Monotonic across all ten deciles, which is what licenses shipping the disagreement band as an error indicator. This measured curve is the expected-error lookup in the released tool.

Appendix E. Correction attempt history and ship decisions

Attempt	Form	Held-out result	Decision
Transonic F1 to F3, null	power laws in $(M - M_{crit})$	Harris 78 percent worse; TN 1546 only 9.3 percent better	Rejected by pre-registered one-shot
Transonic similarity S1, S2	thickness-scaled similarity	86.0 vs 89.7 counts cross-validated; S2 overfits	Rejected, exploratory
Lift-break onset shift	one parameter	+2 percent, worsens TN 1546	Rejected
Lift-break alpha-detached onset	one parameter	-2 percent	Rejected
Drag correction v1	21-term ridge	+10.5 percent, both transfers improve	Shipped, later superseded

Attempt	Form	Held-out result	Decision
Drag correction v2	16-feature boosted trees	+15.7 percent but one facility transfer worsens	Rejected
Drag correction v3	9-feature boosted trees, tighter regularization	+12.4 percent, both transfers improve	Shipped
Lift correction v2	16-feature boosted trees	+26.7 percent, both transfers improve	Shipped

The ship rule was declared before each fit: improvement on airfoil-disjoint folds AND in both cross-facility transfer directions. Eight attempts, two survivors and one superseded predecessor. The rejections are as much a result as the ships.

Appendix F. Reproducibility

Environment: NeuralFoil 0.3.3, AeroSandbox 4.2.10, numpy 2.5.2, scipy, scikit-learn 1.9.0, Python 3.14.4, pinned in a dedicated virtual environment; XFOIL 6.99 built headless from source. The browser port of the network and of both correction models is verified against the Python path on every page load (40 reference runs to about 1e-15; tree ensembles to 0.0 exactly).

Artifact	Contents
lsat-corpus.csv	Parsed measured corpus, 14,773 points with source and configuration
lsat-nf.csv, lsat-nf2.csv	All eight model sizes at every measured condition, base and extended captures
lsat-xfoil.csv	Every XFOIL run, 7,897 converged points
lsat-report.txt, lsat-lift-report.txt	Error maps for drag and lift, worst and best airfoils
lsat-headtohead2.txt	Final four-way comparison output
correction-cd3.json, correction-cl2.json	Shipped correction models with reference vectors
oof2.csv, oof3.csv	Out-of-fold correction predictions used for scoring
master-dataset.csv	The 92-point primary transonic dataset with provenance and uncertainty
research-answer.md	This document's parts 1 to 13 in source form
lsat_*.py, atlas*.py, fit_definitive.py, probes.py	Every pipeline script, in execution order

All artifacts are published under study/ at kaanboge.github.io/neuralfoil. Data files carry the licenses of their sources: the UIUC corpus is GPL v2 (Selig et al.), the NACA and NASA reports are public domain, and NeuralFoil is MIT (Sharpe).